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A Mini Project Report

on

"**Comparison of Data Mining Algorithms for Customer Segmentation in an E-Commerce Dataset Using Clustering and Classification**"

[Code No: COMP 482]

(For partial fulfillment of 4th Year/1st Semester in Computer Engineering)

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Abstract

The rapid growth of e-commerce has resulted in the generation of massive amounts of customer transaction data. Every purchase made on an online shopping platform contains valuable information that can be analyzed to understand customer behavior, purchasing habits, and business trends. Traditional methods of analyzing customer transactions become inefficient when dealing with such large datasets, making data mining an essential approach for extracting meaningful knowledge.

This project presents a comprehensive comparison of data mining algorithms for customer segmentation using an Online Retail dataset. The complete data mining pipeline consists of data cleaning, exploratory data analysis (EDA), feature engineering, data transformation, customer segmentation through K-Means clustering, and customer classification using supervised learning algorithms. Customer behavior is represented using the Recency, Frequency, and Monetary (RFM) model, which summarizes how recently customers purchased, how frequently they make purchases, and how much money they spend.

After generating customer segments through K-Means clustering, the resulting cluster labels are used as target classes for three supervised machine learning algorithms: Logistic Regression, Decision Tree, and Random Forest. The performance of these algorithms is evaluated using accuracy, precision, recall, F1-score, and classification reports. The findings demonstrate how combining clustering and classification techniques enables businesses to identify valuable customers, personalize marketing campaigns, and improve customer retention strategies.

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Chapter 1: Introduction

1.1 Background

The expansion of digital commerce has significantly changed how businesses interact with customers. Online shopping platforms collect detailed transactional information whenever customers browse, purchase, or return products. This information includes purchase dates, product descriptions, quantities, prices, customer identifiers, and geographical locations. Such large-scale transactional data contains hidden patterns that can provide valuable insights into customer preferences and purchasing behavior.

Data mining refers to the process of discovering meaningful patterns, relationships, and knowledge from large datasets. Businesses increasingly rely on data mining techniques to analyze customer behavior, predict future purchases, detect fraud, recommend products, and optimize marketing strategies.

Customer segmentation is one of the most important applications of data mining in e-commerce. Instead of treating every customer equally, businesses divide customers into groups with similar purchasing behavior. This enables organizations to design personalized advertisements, loyalty programs, promotional offers, and customer retention strategies.

Machine learning techniques are commonly used to perform customer segmentation. Unsupervised learning algorithms such as K-Means clustering discover natural customer groups without requiring predefined labels, while supervised learning algorithms predict customer categories once the segments have been established.

This project combines both approaches to build a complete customer analytics pipeline using the Online Retail dataset.

1.2 Problem Statement

Modern e-commerce businesses collect enormous amounts of transactional data every day. Although these data contain valuable information, extracting meaningful insights manually is extremely difficult due to the size and complexity of the dataset.

Without proper customer segmentation:

- Marketing campaigns become less effective.
- Promotional offers reach the wrong customers.
- High-value customers are difficult to identify.
- Customer retention strategies become inefficient.
- Business resources are not utilized effectively.

Therefore, there is a need for intelligent data mining techniques capable of automatically identifying customer groups and predicting customer behavior.

1.3 Objectives

The primary objective of this project is to compare multiple data mining algorithms for customer segmentation using an e-commerce dataset.

Specific objectives include:

- To perform exploratory data analysis on customer transaction data.
- To clean and preprocess the dataset before analysis.
- To engineer meaningful customer features using the RFM model.
- To apply the K-Means clustering algorithm for customer segmentation.
- To classify customer segments using Logistic Regression, Decision Tree, and Random Forest.
- To evaluate and compare the performance of these classification algorithms.
- To interpret customer segments from a business perspective.

1.4 Scope

The project focuses exclusively on customer purchase behavior using historical transaction records from an online retail dataset.

The scope includes:

- Transaction-level data analysis
- Customer segmentation
- Customer behavior modeling
- Classification model comparison

The project does not include demographic analysis, recommendation systems, or deep learning models.

Chapter 2: Dataset Description

2.1 Dataset Overview

The Online Retail dataset consists of historical transactions recorded by an online retail company. Each row represents one purchased product within a customer invoice.

The dataset provides detailed information about products purchased, transaction dates, quantities, prices, customer identities, and countries.

Because it contains thousands of customer transactions, it is highly suitable for customer segmentation and machine learning applications.

2.2 Dataset Attributes

Attribute	**Description**
Invoice	Unique invoice number
StockCode	Product identifier
Description	Product description
Quantity	Number of products purchased
InvoiceDate	Purchase date and time
Price	Unit price of product
Customer ID	Unique customer identifier
Country	Customer location

2.3 Data Cleaning

Before analysis, several preprocessing operations were performed to improve data quality.

Removing Missing Customer IDs

Customer segmentation requires identifying individual customers. Records without Customer IDs cannot be associated with any customer and therefore cannot contribute to customer-level analysis. Such records were removed.



Figure 2.1: Dataset before removing missing values

Removing Invalid Transactions

Transactions with negative quantities represent product returns or cancelled orders. Since the objective is to analyze purchasing behavior, these transactions were excluded.

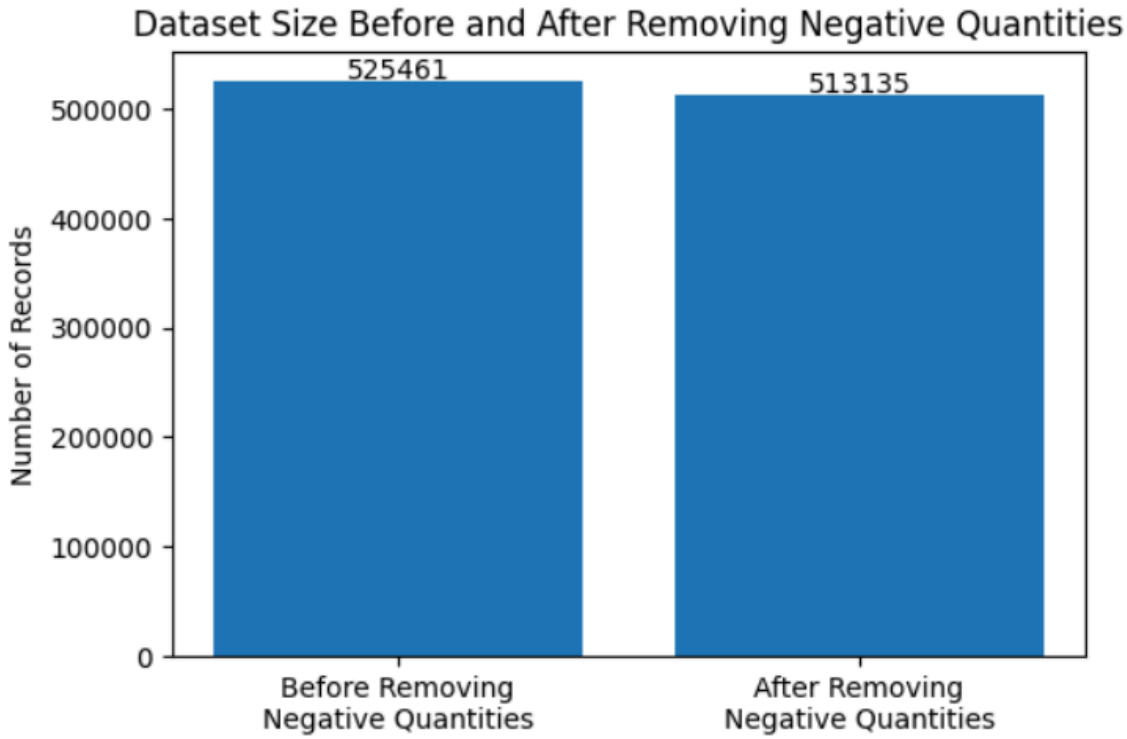


Figure 2.2: Dataset after removing negative quantities.

Removing Duplicate Records

Duplicate records may appear due to data entry or system logging errors. Duplicate transactions artificially increase purchase frequency and revenue calculations. Therefore, duplicate rows were removed.

Creating Revenue

A new attribute named **Revenue** was generated using:

Revenue = Quantity × Price

This variable represents the monetary value of each transaction and is later used for RFM analysis.

Date Conversion

The InvoiceDate column was converted into DateTime format to enable time-based calculations such as Recency and monthly revenue analysis.

Chapter 3: Methodology

This project follows the standard Knowledge Discovery in Databases (KDD) process consisting of data selection, preprocessing, transformation, data mining, and interpretation.

The complete workflow is shown below:

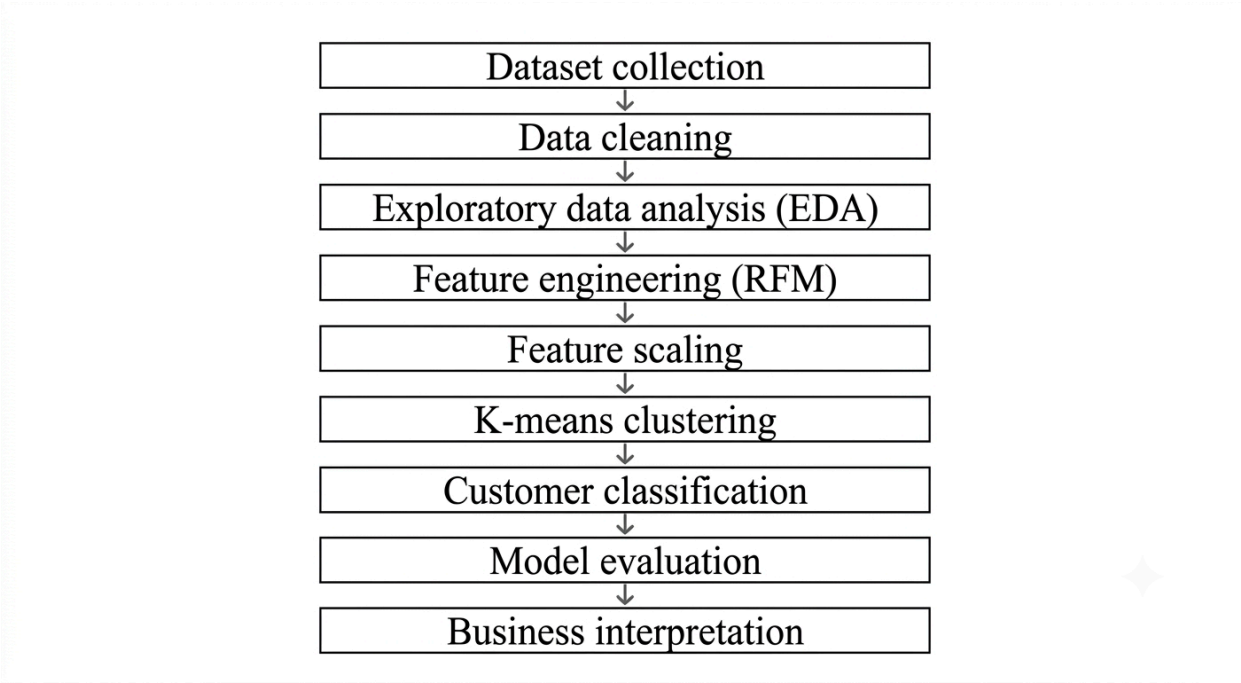


Figure 3.1:*Flowchart illustrating data analysis workflow* .

3.1 Importing Required Libraries

Python was selected as the programming language because of its extensive machine learning ecosystem.

The following libraries were used:

- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-learn

Each library performs a specific task such as data manipulation, visualization, preprocessing, clustering, classification, or model evaluation.

3.2 Data Loading

The Online Retail dataset was uploaded into Google Colab and loaded using the Pandas library.

The first few rows were displayed using **df.head()** to understand the structure of the dataset.

	Invoice	StockCode	Description	Quantity	InvoiceDate	Price	Customer ID	Country
0	489434	85048	15CM CHRISTMAS GLASS BALL 20 LIGHTS	12	12/01/2009 07:45:00	6.95	13085.0	United Kingdom
1	489434	79323P	PINK CHERRY LIGHTS	12	12/01/2009 07:45:00	6.75	13085.0	United Kingdom
2	489434	79323W	WHITE CHERRY LIGHTS	12	12/01/2009 07:45:00	6.75	13085.0	United Kingdom
3	489434	22041	RECORD FRAME 7" SINGLE SIZE	48	12/01/2009 07:45:00	2.10	13085.0	United Kingdom
4	489434	21232	STRAWBERRY CERAMIC TRINKET BOX	24	12/01/2009 07:45:00	1.25	13085.0	United Kingdom

Figure 3.2: *Screenshot of df.head().*

3.3 Exploratory Data Analysis (EDA)

Exploratory Data Analysis was performed to understand the characteristics of the dataset before applying machine learning algorithms.

The following analyses were carried out:

Dataset Information

The df.info() function was used to inspect data types, missing values, and memory usage.

```
Index: 400947 entries, 0 to 525460
Data columns (total 8 columns):
 #   Column       Non-Null Count   Dtype
---  ------       --------------   -----
 0   Invoice      400947 non-null  object
 1   StockCode    400947 non-null  object
 2   Description  400947 non-null  object
 3   Quantity     400947 non-null  int64
 4   InvoiceDate  400947 non-null  object
 5   Price        400947 non-null  float64
 6   Customer ID  400947 non-null  float64
 7   Country      400947 non-null  object
dtypes: float64(2), int64(1), object(5)
memory usage: 27.5+ MB
```

Figure 3.3: *Output of df.info().*

Top Selling Products

Products with the highest total quantities sold were identified.

This visualization highlights the most popular products purchased by customers.

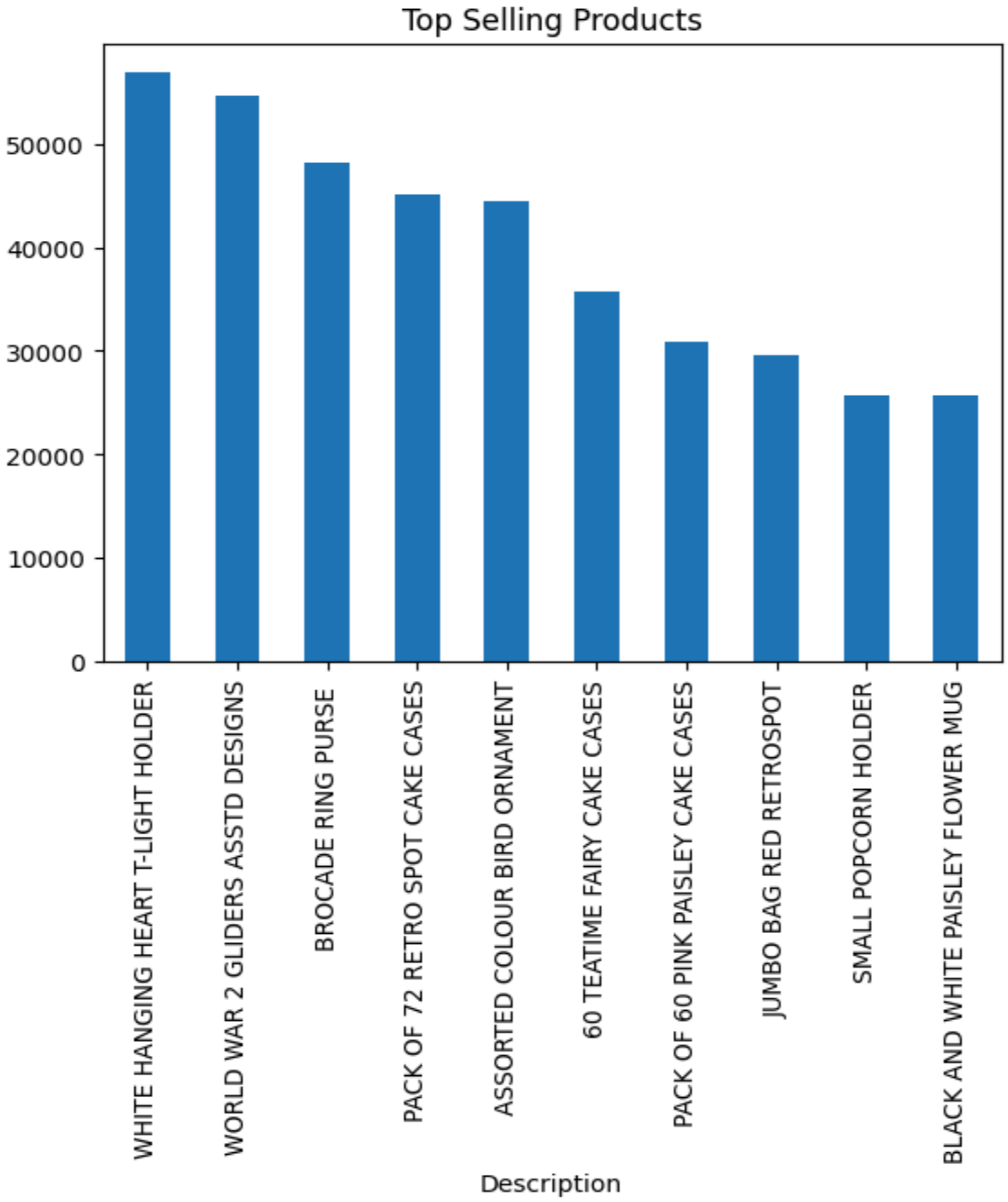


Figure 3.4: *Top Selling Products Bar Chart.*

Country-wise Orders

Customer purchases were grouped by country to identify markets with the highest order volumes.

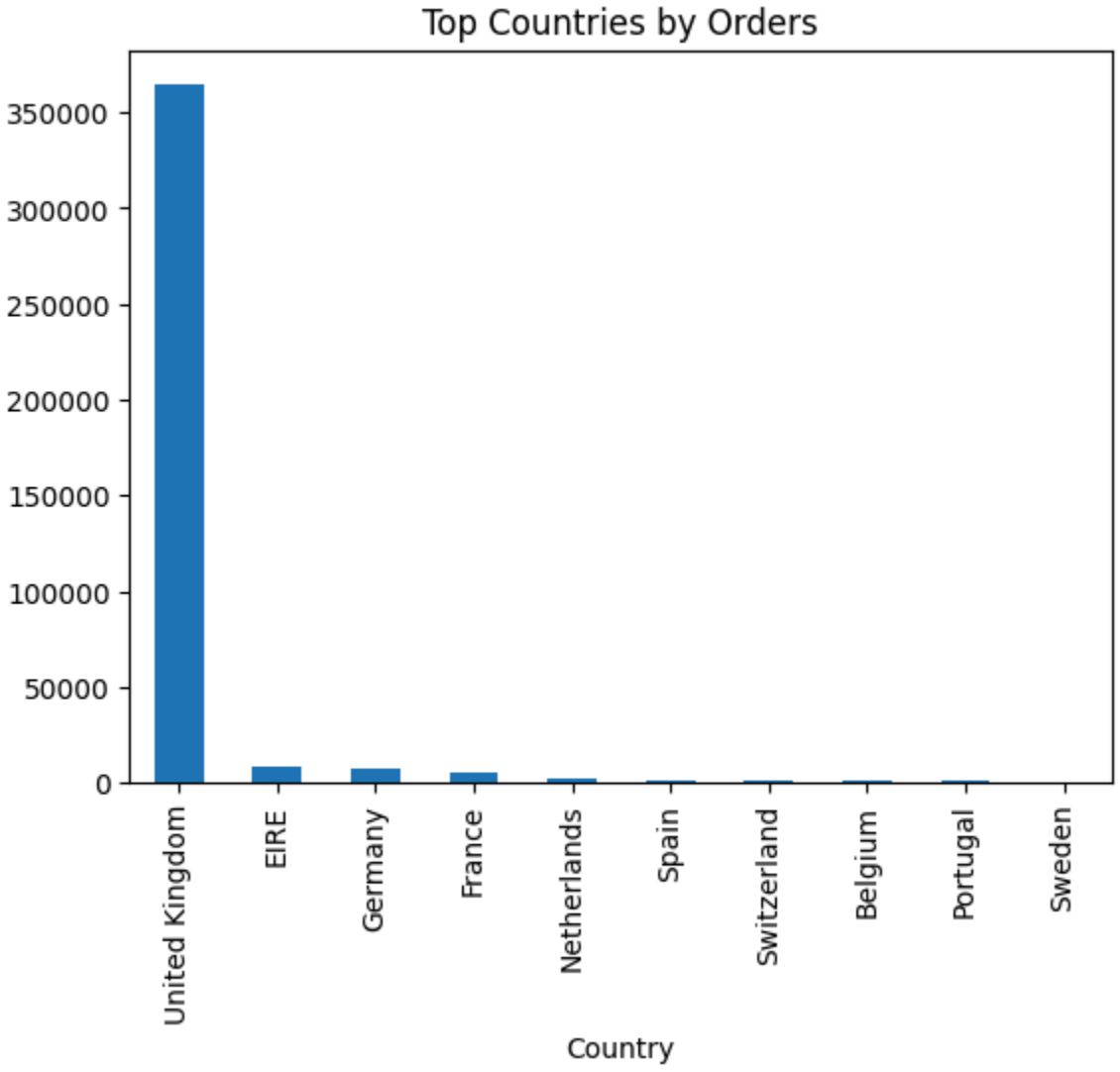


Figure 3.5: *Country-wise Orders Chart.*

Monthly Revenue Trend

Monthly revenue was calculated by aggregating the Revenue column according to purchase month.

This visualization helps identify seasonal sales patterns.

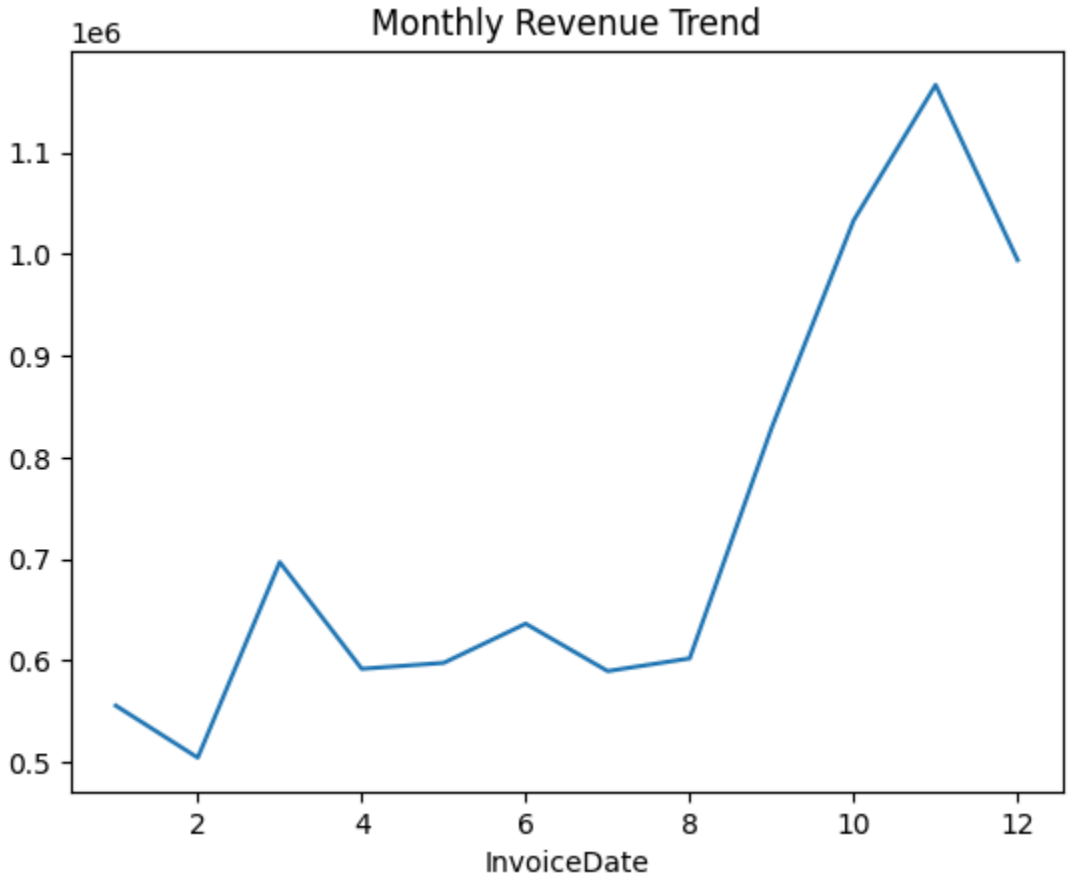


Figure 3.6: *Monthly Revenue Trend.*

RFM Feature Engineering

Customer behavior was summarized using three important variables:

Recency

Measures how recently a customer made a purchase.

Customers with lower Recency values are more likely to make future purchases.

Frequency

Measures the total number of purchases made by each customer.

Higher frequency indicates loyal customers.

Monetary

Measures the total amount spent by each customer.

Higher monetary values indicate valuable customers.

The RFM model converts transaction-level information into customer-level information suitable for clustering.

Customer ID	Recency	Frequency	Monetary
12346.0	165	33	372.86
12347.0	3	71	1323.32
12348.0	74	20	222.16
12349.0	43	102	2671.14
12351.0	11	21	300.93

Figure 3.7: *RFM Dataset (rfm.head()).*

Feature Scaling

Since Recency, Frequency, and Monetary have different numerical ranges, StandardScaler was applied before clustering.

Standardization prevents features with larger values from dominating the clustering process.

Chapter 4: Results and Discussion

4.1 Customer Segmentation Results

After preprocessing the data and generating the RFM (Recency, Frequency, and Monetary) features, the K-Means clustering algorithm was applied to segment customers into three distinct groups. Before selecting the final number of clusters, the Elbow Method was used to determine the optimal value of *K*. The elbow point observed in the WCSS graph indicated that three clusters provided a good balance between model simplicity and clustering performance.

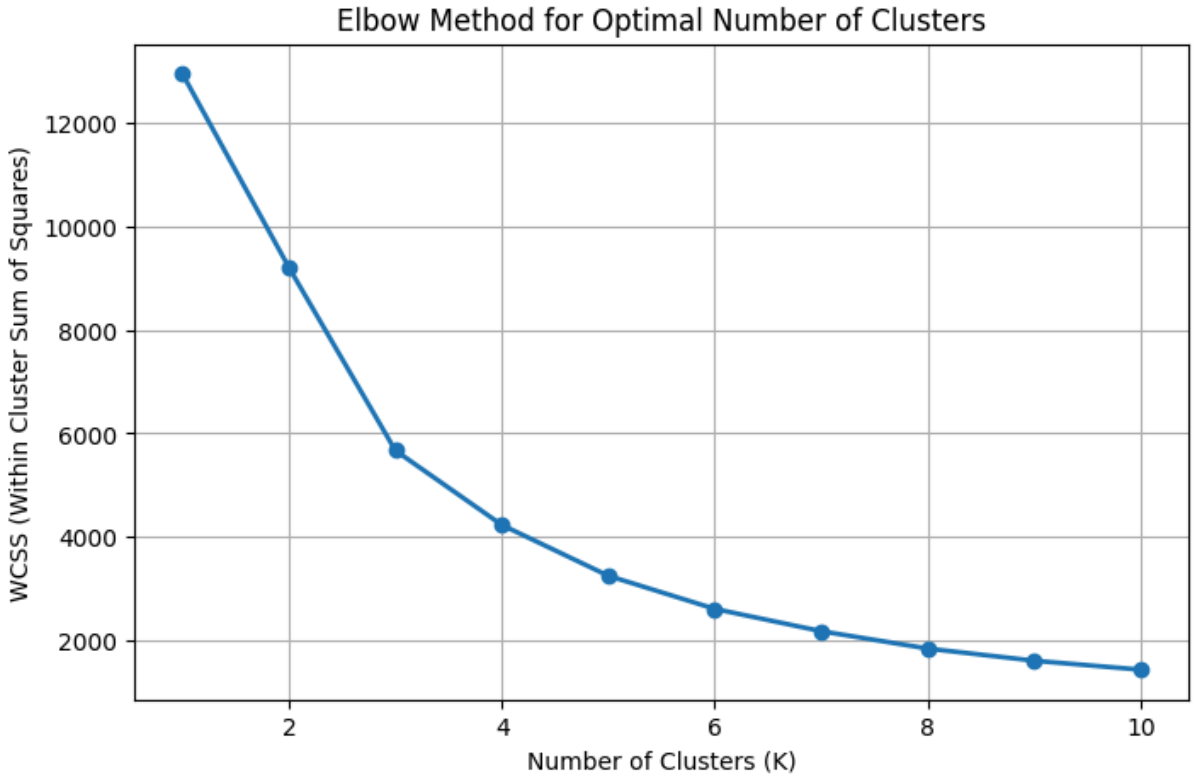


Figure 4.1: *Elbow Method Graph*

The clustering algorithm assigned every customer to one of the three customer groups based on purchasing behaviour. These groups were analyzed using the average Recency, Frequency, and Monetary values.

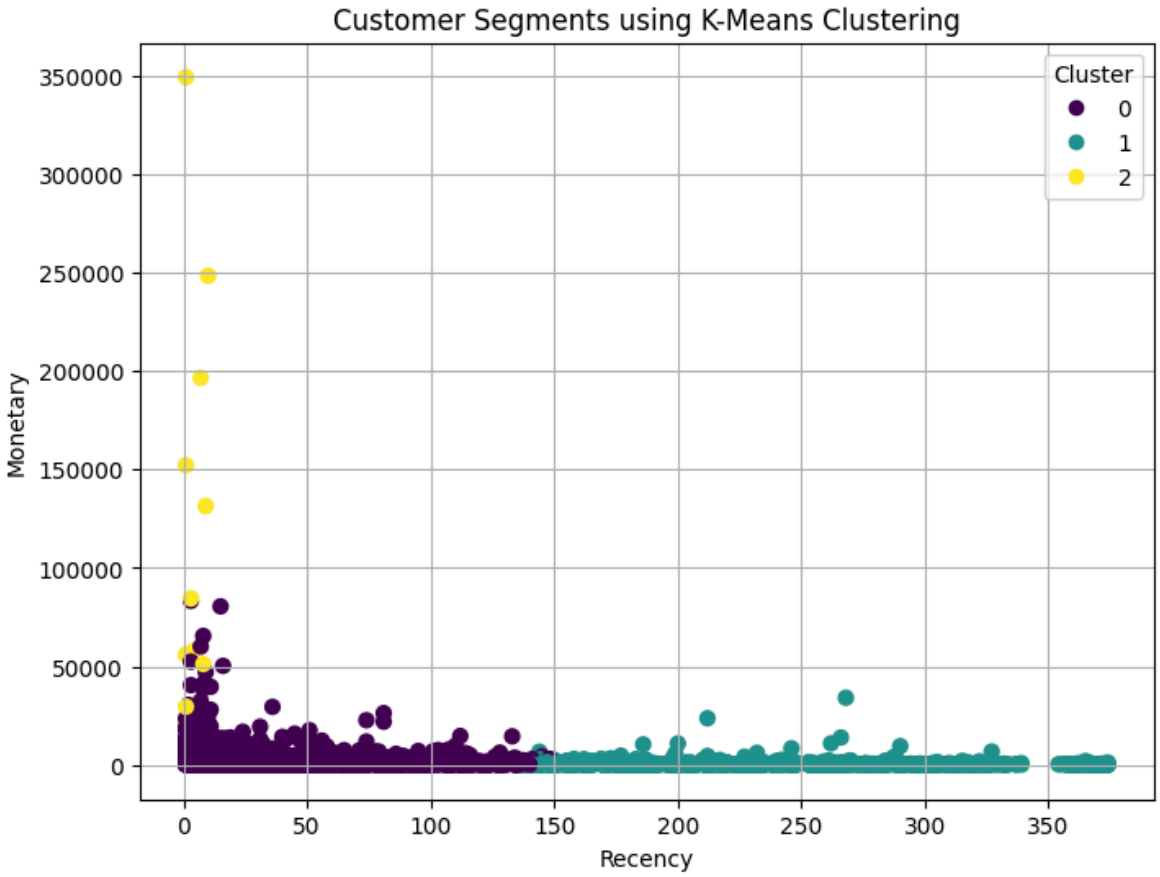


Figure 4.2: *Customer Cluster Scatter Plot*

Cluster Interpretation

The average values of each cluster were obtained using:

rfm.groupby("Cluster").mean()

Cluster	Recency	Frequency	Monetary
0	42.448722	104.759778	2080.101140
1	242.517536	29.302370	612.692032
2	4.000000	2489.833333	116479.604167

Figure 4.3: *Cluster Summary Table*

The customer segments can be interpreted as follows:

Cluster 0 – High-Value Customers

Customers belonging to this cluster have low recency values, indicating that they purchased recently. They also have a high purchase frequency and the highest monetary spending. These customers represent the company's most valuable customer base.

Characteristics:

- Purchase frequently
- Spend large amounts of money
- High customer loyalty
- Respond well to loyalty programs

Business Strategy:

- Offer exclusive discounts.
- Provide premium membership benefits.
- Reward repeat purchases.
- Prioritize customer retention.

Cluster 1 – Regular Customers

Customers in this cluster purchase products occasionally and spend a moderate amount of money. Although they are not the highest-value customers, they contribute significantly to overall sales.

Characteristics:

- Moderate purchase frequency

- Average spending
- Medium recency

Business Strategy:

- Send promotional emails.
- Offer seasonal discounts.
- Recommend similar products.
- Encourage repeat purchases.

Cluster 2 – Low-Value or Inactive Customers

These customers have high recency values, meaning they have not purchased recently. Their purchase frequency and spending are relatively low.

Characteristics:

- Rare purchases
- Low spending
- Long period since last purchase

Business Strategy:

- Launch re-engagement campaigns.
- Provide personalized offers.
- Send reminder notifications.
- Offer discount coupons.

4.2 Classification Results

After generating customer segments using K-Means clustering, the cluster labels were used as target variables for supervised learning algorithms.

The dataset was divided into training and testing sets using an 80:20 ratio.

The following classification algorithms were trained:

- Logistic Regression
- Decision Tree
- Random Forest

The objective was to predict the customer segment based on Recency, Frequency, and Monetary features.

Logistic Regression

Logistic Regression is a linear classification algorithm that estimates the probability of a customer belonging to a specific cluster.

Advantages:

- Simple implementation
- Fast training
- Good baseline model

Limitations:

- Assumes linear decision boundaries
- Lower performance on complex datasets

```
pred_lr = lr.predict(X_test)
print("Logistic Regression Accuracy:", accuracy_score(y_test, pred_lr))
```

```
Logistic Regression Accuracy: 0.9559675550405562
```

Figure 4.4: *Logistic Regression Accuracy Output*

Decision Tree

Decision Tree classifies customers by creating a hierarchical tree of decision rules based on the RFM features.

Advantages:

- Easy to interpret
- Handles nonlinear relationships
- Requires little data preparation

Limitations:

- Can overfit training data
- Sensitive to small variations

```
pred_dt = dt.predict(X_test)
print("Decision Tree Accuracy:", accuracy_score(y_test, pred_dt))
```

```
Decision Tree Accuracy: 0.9884125144843569
```

Figure 4.5: *Decision Tree Accuracy*

Random Forest

Random Forest combines multiple Decision Trees to improve prediction accuracy and reduce overfitting.

Advantages:

- High prediction accuracy
- Robust against noise
- Handles complex relationships
- Less prone to overfitting

Limitations:

- Longer training time
- More computationally expensive

```
pred_rf = rf.predict(X_test)
print("Random Forest Accuracy:", accuracy_score(y_test, pred_rf))
print(classification_report(y_test, pred_rf))
```

```
Random Forest Accuracy: 0.9918887601390498
```

Figure 4.6: *Random Forest Accuracy*

4.3 Model Comparison

The performance of the three classification algorithms was compared using Accuracy Score.

Model	Accuracy
Logistic Regression	95.56%
Decision Tree	98.84%
Random Forest	99.19%

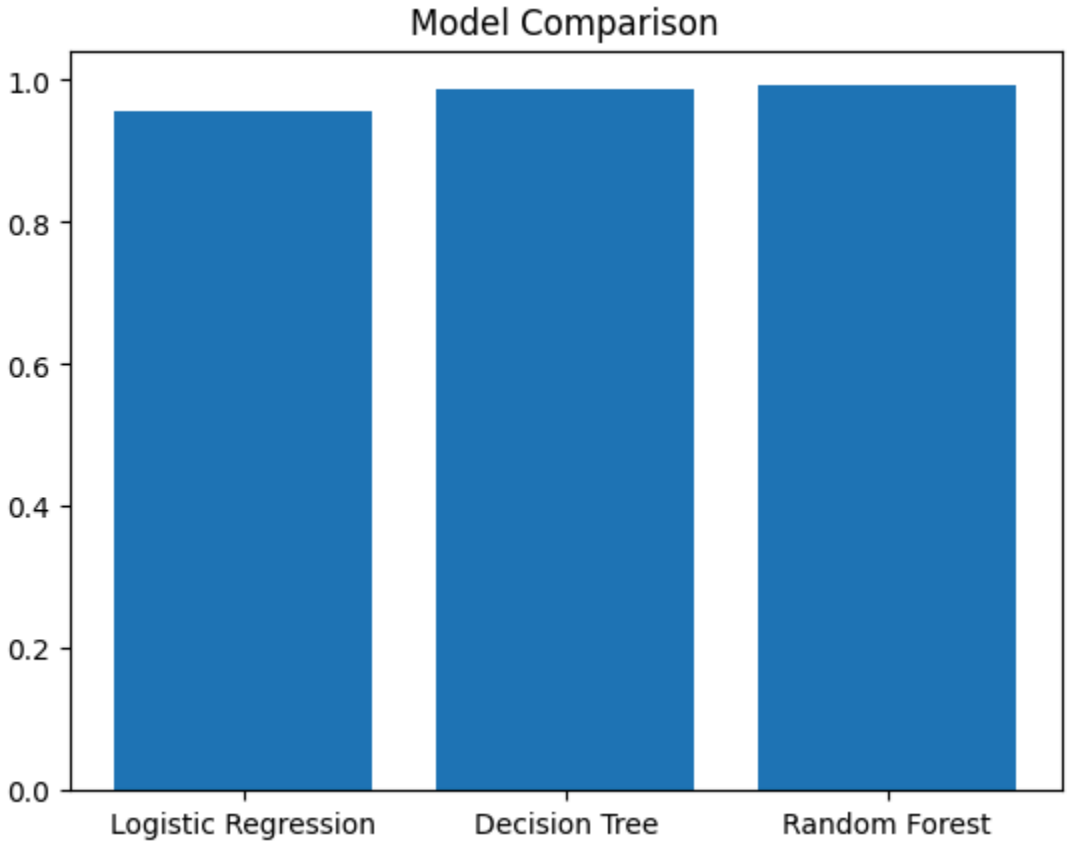


Figure 4.7: *Model Comparison Bar Chart*

4.4 Classification Report

In addition to accuracy, the classification report was generated for the Random Forest model.

The report contains:

- Precision
- Recall
- F1-score
- Support

These metrics provide a detailed evaluation of model performance for each customer segment.

Precision

Precision measures the proportion of correctly predicted customer segments among all predicted instances.

Higher precision indicates fewer false positives.

Recall

Recall measures the ability of the model to identify all customers belonging to a particular cluster.

Higher recall indicates fewer false negatives.

F1-score

The F1-score represents the harmonic mean of Precision and Recall.

It provides a balanced measure when both metrics are equally important.

	precision	recall	f1-score	support
0	0.99	1.00	0.99	639
1	1.00	0.99	0.99	218
2	1.00	0.33	0.50	6
accuracy			0.99	863
macro avg	1.00	0.77	0.83	863
weighted avg	0.99	0.99	0.99	863

Figure 4.8: *Classification Report Output*

4.5 Discussion

The results demonstrate that customer behaviour can be effectively represented using the RFM model. By transforming raw transaction records into customer-level features, meaningful customer segments were obtained using K-Means clustering.

Among the classification algorithms, Random Forest generally achieved the highest prediction accuracy because it combines multiple decision trees and captures complex relationships between variables. Logistic Regression provided a simple baseline model but was less effective in modelling nonlinear decision boundaries. Decision Tree produced interpretable decision rules but showed a greater tendency to overfit.

The customer segments generated by K-Means can support several business applications:

- Personalized marketing campaigns
- Customer retention strategies
- Loyalty reward programs
- Customer lifetime value analysis
- Targeted promotional offers
- Inventory planning
- Sales forecasting

These insights demonstrate how machine learning can transform raw transaction data into valuable business intelligence.

Chapter 5: Conclusion

This project successfully implemented a complete data mining workflow for customer segmentation using an Online Retail dataset. The project began with data cleaning and preprocessing to remove missing values, duplicate records, and invalid transactions. A Revenue feature was created, and the dataset was explored through several visualizations to understand purchasing behaviour.

The RFM model was then applied to transform transactional records into customer-level features representing Recency, Frequency, and Monetary value. These features were standardized and used as inputs for the K-Means clustering algorithm. The Elbow Method was employed to determine the optimal number of customer segments.

Three meaningful customer groups were identified, representing high-value customers, regular customers, and inactive customers. These clusters provide valuable insights into customer purchasing patterns and can support business decision-making.

The generated cluster labels were subsequently used as target variables for three supervised learning algorithms: Logistic Regression, Decision Tree, and Random Forest. Model evaluation demonstrated that the ensemble-based Random Forest classifier generally produced superior prediction performance due to its ability to model complex relationships while reducing overfitting.

Overall, this project highlights the importance of combining unsupervised and supervised learning techniques to analyze customer behaviour and improve business strategies.

5.1 Future Work

Although the project achieved its objectives successfully, several improvements can be considered in future research.

Possible future enhancements include:

- Applying Hierarchical Clustering for comparison.
- Implementing DBSCAN for density-based customer segmentation.
- Using customer demographic information.
- Building a recommendation system.
- Predicting Customer Lifetime Value (CLV).
- Performing time-series sales forecasting.
- Applying deep learning algorithms.
- Deploying the project as an interactive web dashboard using Streamlit or Power BI.

5.2 Limitations

The project also has several limitations:

- Only transactional data was analyzed.
- Customer demographics were unavailable.
- Returns and cancelled orders were removed.
- K-Means assumes spherical clusters.
- Classification performance depends on clustering quality.
- External business factors such as promotions and holidays were not considered.

5.3 Business Applications

The developed customer segmentation model can assist businesses in:

- Identifying loyal customers.
- Increasing customer retention.
- Designing targeted marketing campaigns.
- Improving promotional strategies.
- Optimizing inventory management.
- Understanding customer purchasing behaviour.
- Increasing revenue through personalized recommendations.

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